

Donovan Reynolds

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SUMMARY

Graduate in computer engineering with interests in embedded systems, system software, and software engineering. Experienced with low-level programming languages, computer architecture, and systems development through academic projects and research. I completed my Master's in Computer Engineering at the University of Central Florida (UCF).

WORK EXPERIENCE

Graduate Instructor Systems Software: UCF

August 2025 - Current

Skills: *C, C++, Bash, Linux, Antlr, Compilers, Git*

- Led lab sections covering Linux system calls, shell development, and compiler design for undergraduate students.
- Guided students through the implementation of a custom shell and basic compiler projects using C/C++.
- Mentored students in debugging, code optimization, and software development best practices in Linux environments.
- Reinforced core systems programming concepts through hands-on instruction and technical support.

Undergraduate TA Embedded Systems: UCF

August 2023 - August 2024

Skills: *C, MSP430, Code Composer Studio*

- Facilitated lab sessions teaching embedded systems concepts using the MSP430 microcontroller platform.
- Supported students in developing and debugging C programs using JTAG and Code Composer Studio.
- Interpreted technical documentation to troubleshoot hardware/software issues and improve lab efficiency.
- Provided one-on-one technical guidance to strengthen student understanding of low-level programming concepts.

High School Math Teacher: A.C.E.

August 2023 - May 2025

Skills: *Communication, Adaptability, Problem-Solving*

- Designed and taught introductory Python programming curriculum for high school students.
- Delivered instruction in mathematics ranging from Pre-Algebra through Pre-Calculus to diverse learning levels.
- Adapted teaching strategies to improve student comprehension, engagement, and academic performance.
- Built strong classroom communication and problem-solving skills through individualized student support.

Lead Instructor: Mathnasium of Lake Mary

Feb 2020 - August 2023

Skills: *Leadership, Collaboration, Conflict Resolution*

- Led and trained a team of instructors to effectively teach students across a wide range of math skill levels.
- Coordinated staff development and instructional strategies to improve student learning outcomes.
- Resolved scheduling, instructional, and student engagement challenges in a fast-paced tutoring environment.
- Fostered a collaborative team culture focused on high-quality instruction and customer satisfaction.

PROJECTS

Arduscript [Link to Demo](#)

Skills: *Python, Antlr, Compilers*

- Designed a custom compiler in Python
- Created a custom scripting language that is parsed by the Antlr toolchain.
- Compiles into Arduino IDE compatible code to be flashed to any compatible system

B.I.R.D.S. [Link to Demo](#)

Skills: *Python, C++, ESP32, ATMEGA, Jetson Nano, PCB Design, Machine Learning, Image Processing*

- Designed a system to aid in training birds to pick up cigarettes
- Wrote code in C++ to go through and process various sensors and trigger the next stages of the system
- Developed an image recognition model to detect cigarettes using the Jetson Nano
- Wrote code in Python to interface the image model and send data to an embedded device

EDUCATION

2024 - 2026	Master's Degree (Computer Engineering)	University of Central Florida	(GPA: 4.0/4.0)
2020 - 2024	Bachelor's Degree (Computer Engineering)	University of Central Florida	(GPA: 3.9/4.0)